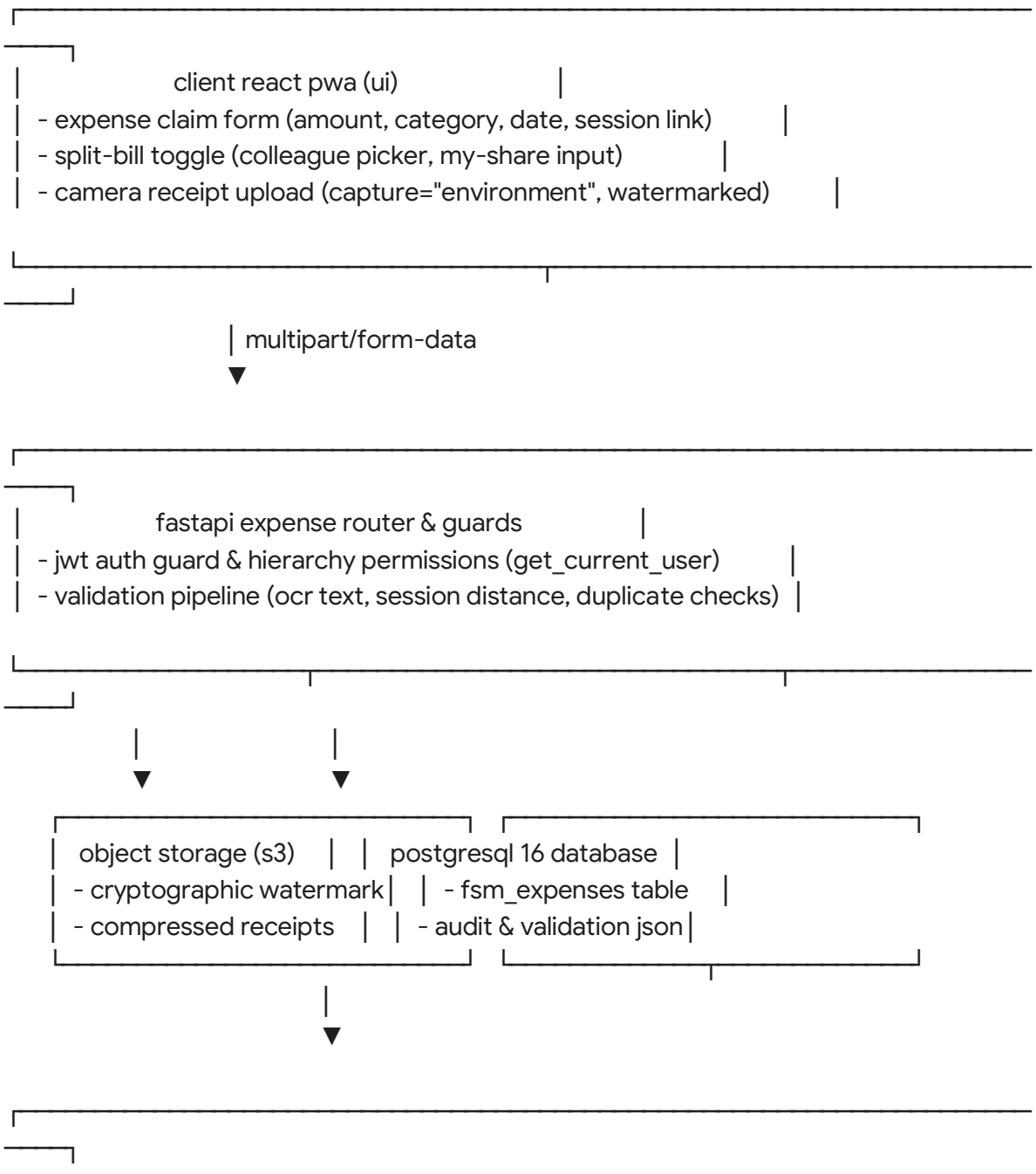
expense management module: end-to-end architecture & step-by-step flow

1. high-level module architecture



```
|                  manager review & audit dashboard                  |
|  - team claims list (scoped via recursive cte hierarchy_service)   |
|  - automated validation badges (green ok / amber split / red duplicate) |
|  - one-click approve, reject with notes, or mark reimbursed        |

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```

2. database layer specification

primary schema (fsm_expenses)

```
CREATE TYPE expense_category AS ENUM (
    'fuel',
    'travel_ticket',
    'food_allowance',
    'accommodation',
    'toll_parking',
    'client_entertainment',
    'miscellaneous'
);

CREATE TYPE expense_status AS ENUM (
    'submitted',
    'approved',
    'rejected',
    'reimbursed'
);

CREATE TABLE fsm_expenses (
    id UUID DEFAULT gen_random_uuid() PRIMARY KEY,
    employee_id VARCHAR(255) NOT NULL REFERENCES pharma_hrms_employees(id),
    field_session_id VARCHAR(255) REFERENCES pharma_hrms_field_sessions(id) ON DELETE
SET NULL,
    category expense_category NOT NULL,
    amount NUMERIC(10, 2) NOT NULL CHECK (amount    0),
    claim_date DATE NOT NULL,
```

```
    receipt_url TEXT,
    description TEXT,

    -- split-bill metadata
    is_split_bill BOOLEAN DEFAULT FALSE,
    split_with_employee_id VARCHAR(255) REFERENCES pharma_hrms_employees(id),
    receipt_invoice_number VARCHAR(100),
    total_receipt_amount NUMERIC(10, 2),

    -- automated validation & fraud checks
    image_sha256 VARCHAR(64),
    validation_status VARCHAR(50) DEFAULT 'passed', -- 'passed', 'flagged', 'suspicious'
    validation_flags JSONB DEFAULT '[]'::jsonb,

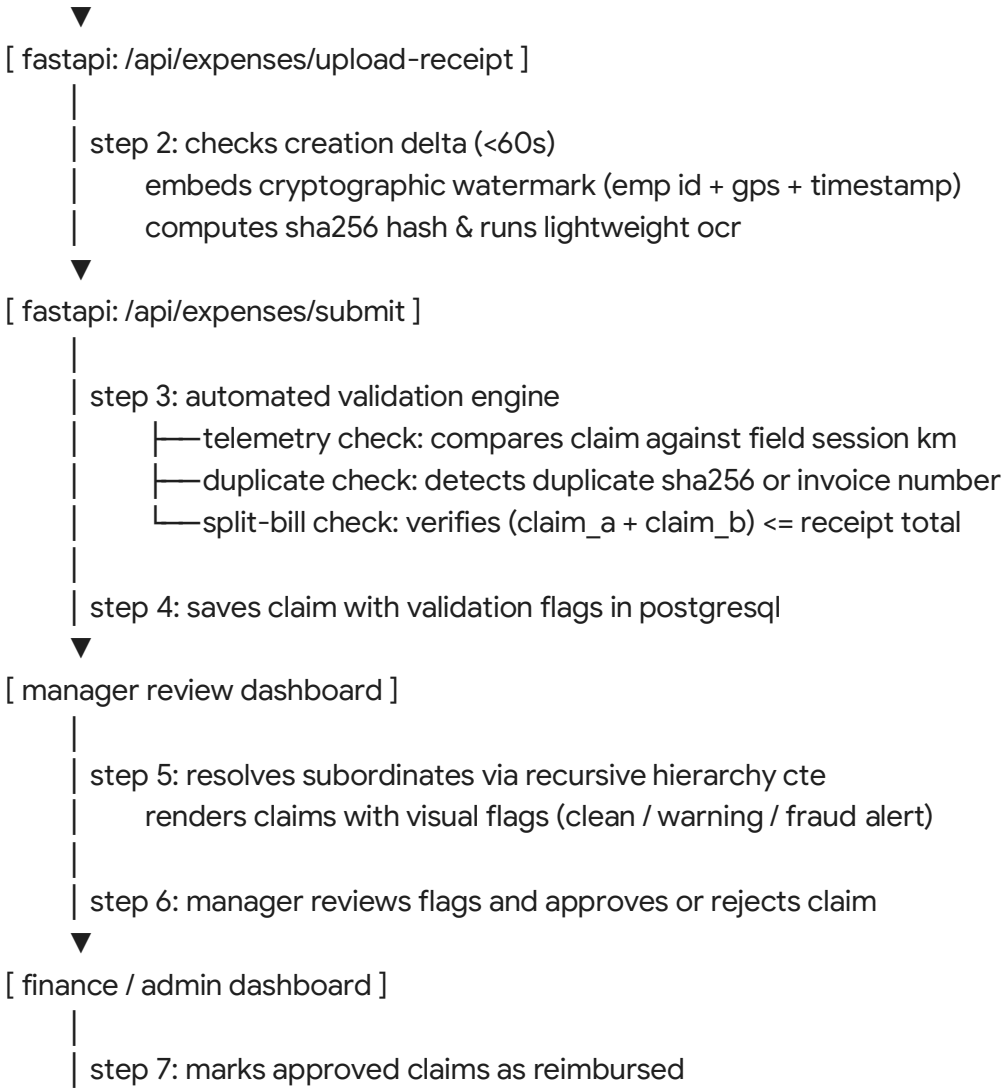
    -- approval workflow
    status expense_status DEFAULT 'submitted',
    reviewed_by VARCHAR(255) REFERENCES pharma_hrms_employees(id),
    reviewed_at TIMESTAMP WITH TIME ZONE,
    review_note TEXT,
    reimbursed_at TIMESTAMP WITH TIME ZONE,

    created_at TIMESTAMP WITH TIME ZONE DEFAULT CURRENT_TIMESTAMP,
    updated_at TIMESTAMP WITH TIME ZONE DEFAULT CURRENT_TIMESTAMP
);

CREATE INDEX idx_expenses_emp_date ON fsm_expenses(employee_id, claim_date);
CREATE INDEX idx_expenses_status ON fsm_expenses(status);
CREATE INDEX idx_expenses_invoice ON fsm_expenses(receipt_invoice_number,
claim_date);
CREATE INDEX idx_expenses_sha ON fsm_expenses(image_sha256);
```

3. step-by-step execution flow

```
[ employee mobile pwa ]
        |
        | step 1: captures receipt photo & inputs details
        |       (optional: flags bill as split with colleague)
```



4. detailed technical implementation steps

step 1: client submission with receipt capture (react pwa)

the executive opens the expense claim form in the pwa:

- selects category (fuel, food, accommodation, travel).
- enters claim amount and transaction date.
- attaches the active field_session_id to correlate against day travel.
- snaps a live receipt using the device camera (capture="environment").
- if dining with a team member, toggles **split bill**, enters their colleague's name, and types the total bill value versus their individual share.

step 2: upload, forensic watermarking & ocr extraction (backend)

the backend processes the raw image stream:

1. **tamper verification:** verifies timestamp delta ($\Delta t < 60s$) to avoid gallery uploads.
2. **cryptographic watermarking:** stamps employee id, server timestamp, and gps

coordinates onto the photo canvas before storage.

3. **hash generation:** computes image_sha256 of the binary stream.
4. **ocr extraction:** parses the image to extract printed dates, bill/invoice numbers, and printed totals.
5. uploads the processed image to object storage and returns the permanent url.

step 3: multi-layered automated validation service

before persisting to fsm_expenses, the backend evaluates three business rules:

- **check A (session distance correlation):** if category is fuel, the service fetches pharma_hrms_field_sessions for that session id. if the logged session distance is 15 km, but fuel claimed is ₹1,200, it appends:
  JSON
  `"code": "DISTANCE_COST_MISMATCH", "severity": "warning", "message": "claimed ₹1,200 for 15 km recorded distance"`

- **check B (shared split-bill evaluation):**
  the service queries existing claims matching the same receipt_invoice_number and date:
  - if current_claim + existing_colleague_claim <= total_receipt_amount:
    it records a green informational flag:
    JSON
    `"code": "SHARED_BILL_LEGITIMATE", "severity": "info", "message": "split bill verified with colleague"`

  - if current_claim + existing_colleague_claim > total_receipt_amount:
    it records a red critical flag:
    JSON
    `"code": "OVERCLAIMED_DUPLICATE_BILL", "severity": "fraud", "message": "combined claims exceed bill total"`

- **check C (duplicate receipt detection):**
  if a different employee previously used the exact same image_sha256 or invoice number on an un-split claim, it flags:
  JSON
  `"code": "DUPLICATE_RECEIPT_DETECTED", "severity": "fraud", "message": "identical receipt previously uploaded"`

step 4: saving claim state

the record is inserted into fsm_expenses with status = 'submitted' and its populated validation_flags array.

step 5: hierarchy-scoped manager review

when a manager accesses the approval screen:

1. the backend runs the recursive common table expression

(hierarchy_service.get_subordinate_ids) to fetch all employees reporting to that manager[cite: 2].

2. the query filters claims where employee_id IN (subordinates)[cite: 2].
3. the manager ui surfaces claims color-coded by flag severity:
    - **clean claims (green):** distance matches gps data, single bill[cite: 1, 3]. manager uses one-click batch approval.
    - **split claims (blue/amber):** shows both colleague claims linked together so the manager can verify that the two halves add up correctly.
    - **flagged claims (red):** displays distance mismatch or duplicate bill alerts. the manager reviews the route breadcrumbs and either requests clarification or rejects the claim with an admin note.

step 6: final approval & reimbursement transition

- when approved: status transitions to approved, recording reviewed_by and reviewed_at[cite: 3].
- when finance settles payouts: an executive or finance officer hits /api/expenses/{id}/reimburse, transitioning the status to reimbursed with reimbursed_at recorded for accounting audits[cite: 3].